Carlos Michael Jimenez-Muñoz

PhD Candidate in Quantum Chemistry · TU Darmstadt

carlos.jimenez@tu-darmstadt.de **carlosmichaeljimenez.github.io** LinkedIn



Research Summary: Doctoral researcher investigating the electronic structure, spectroscopic parameters, and reactivity of transition-metal complexes via modern wavefunction and multireference quantum chemical methods, with a strong focus on multi-scale modeling and QM/MM molecular dynamics (QM/MM-MD) in complex environments.

RESEARCH EXPERIENCE

Ph.D. in Quantum Chemistry 2022 – Present
Technical University of Darmstadt (TU Darmstadt) Darmstadt, Germany
Advisor: Prof. Dr. Vera Krewald

EDUCATION

Erasmus Mundus - Theoretical Chemistry and Computational Modelling (TCCM) 2021 - 2022
KU Leuven Leuven, Belgium
Transitioned to a PhD programme following a competitive doctoral offer.

B.Sc. in Chemistry - 5 years program 2015 - 2021
Yachay Tech University Urcuquí, Ecuador
Advisor: Prof. Dr. Thibault Terencio

COMPUTATIONAL SKILLS & LANGUAGES

- **Programming & HPC:** Python, Bash, Git, SLURM
- **Languages:** Spanish (Native), English (Fluent), German (B1/B2)

TEACHING & ACADEMIC SERVICE

- **Teaching Assistant (TU Darmstadt, 3 years):** Exercise sessions in Quantum Chemistry and Physical Chemistry.
- **Student Supervision:** Co-supervised 3 B.Sc./M.Sc. research interns in computational transition-metal modeling.
- **Curriculum Innovation:** Participant in 2 Teaching Hackathons (DigiMatCh) focused on digital and interactive chemistry education using Jupyter notebooks.

AWARDS & SUPERCOMPUTING (HPC) ALLOCATIONS

- **Best Poster Presentation - Springer**
  46th International Conference on Coordination Chemistry (ICCC) — 2026
- **Supercomputing Resource Extension Grant (3-Year Allocation)**
  45 million core-hours & 55,000 GPU-hours Facility: Lichtenberg II - 2026
- **Travel Grant (CHITEL 2025)**
  Funded by: Freunde der TU Darmstadt — 2025
- **Research Stay Fellowship - Research stay at CEA, Grenoble with Dr. Ragnar Björnsson**
  Funded by: CRC 1487, Iron upgraded! — 2024

SELECTED PUBLICATIONS

- Diederich, T.M., Wehland, T., Schrodt, M., Kochetov, N., Schnegg, A., **Jimenez-Muñoz, C.M.**, Krewald, V., Ni, L., Salas, N.S., Kramm, U.I., Ballmann, J. and Enders, M. **(2025)**, Chem. Eur. J., 31: e202501474
- Mehmel J., **Jimenez-Muñoz, C.M.**, Rivic F., Krewald V., Schäfer R. **(2025)**, Nanoscale (2025) 17 (7): 3849–3860

SELECTED TALKS

- 7th Quantum (Bio-) Inorganic Chemistry Conference (QBIC VII) - Germany (2025)
  "Multi-scale modelling for an accurate prediction of redox potentials in iron coordination compounds"
- Congress of Theoretical Chemist of Latin Expression (CHITEL) - Belgium (2024)
  "Electronic structure of mononuclear iron phthalocyanine complexes in matrix isolation spectroscopy: experiment challenges theory chemistry"
- **Posters:** 9 contributions (2021-2026), including ICCC 2026 (Denmark), ESPA 2024 (Spain) and STC 2025 (Germany).

Complete list of peer-reviewed articles, preprints and talks available in carlosmichaeljimenez.github.io